

Mathematical Formulation

Samuel OCEN

E MAIL: - samuel.ocen@mmu.ac.ug
samuel.ocen@student.uonbi.ac.ke

MOB: +256(0)775 679 505
+256(0)706 109 393

Real-world Problem Formulation

Example Problem: Product Mix Optimization

- Imagine a company that produces two products, A and B. The goal is to maximize profit, given certain constraints on resources.
- **Decision Variables:**
 x_1 : Quantity of product A to produce.
 x_2 : Quantity of product B to produce.
- **Objective Function (to be maximized):**
 - Maximize $Z=3x_1+2x_2$ Maximize $Z= 3x_1+2x_2$ (Maximizing profit, where the coefficients 3 and 2 represent the profit per unit of products A and B, respectively.)

Constraints

1. **Resource Constraint for Raw Material:** $2x_1 + x_2 \leq 1000$

(The company has 1000 units of raw material available.)

2. **Labor Constraint:** $4x_1 + 2x_2 \leq 2400$

(Labor hours available are limited to 2400.)

3. **Machine Constraint:** $3x_1 + 2x_2 \leq 1500$

(Machine hours available are limited to 1500.)

4. **Non-negativity Constraint:** $x_1, x_2 \geq 0$

(The production quantity cannot be negative.)

This formulation helps represent the real-world problem in terms of decision variables, an objective function, and constraints.

Standard Form of Linear Programming

- The standard form of a linear programming problem is as follows:

Objective Function (to be maximized):

$$\text{Maximize } Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

Constraints:

1. $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$

2. $a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$

3. $\vdots$

4. $a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m$

Non-negativity Constraint

$$x_1, x_2, \dots, x_n \geq 0$$

In this standard form:

- Z is the objective function to be maximized.
- $c_1, c_2, \dots, c_n$ are the coefficients of the decision variables in the objective function.
- a_{ij} represents the coefficients of the decision variables x_j in the i -th constraint.
- $b_1, b_2, \dots, b_m$ are the right-hand side constants in the constraints.

Transforming a real-world problem into standard form allows for the use of various optimization algorithms and techniques.

Manufacturing Optimization Problem

Consider a manufacturing company producing two products, A and B. The goal is to determine the optimal quantity of each product to maximize profit while satisfying resource constraints.

Decision Variables:

Let

- x_1 be the quantity of product A to produce.
- x_2 be the quantity of product B to produce.

Objective Function:

Maximize the profit function:

$$Z = 3x_1 + 2x_2$$

Constraints:

1. Raw Material Constraint:

The raw material available is limited to 1000 units, expressed as:

$$2x_1 + x_2 \leq 1000$$

2. Labor Constraint:

Labor hours are limited to 2400, expressed as:

$$4x_1 + 2x_2 \leq 2400$$

3. Machine Constraint:

Machine hours are limited to 1500, expressed as:

$$3x_1 + 2x_2 \leq 1500$$

4. Non-negativity Constraint:

The production quantity cannot be negative:

$$x_1, x_2 \geq 0$$

Python Code

```
In [1]: #Importing necessary Libraries
        from pulp import LpProblem, LpMaximize, LpVariable

        # Create a linear programming problem
        model = LpProblem(name="Product_Mix_Optimization", sense=LpMaximize)

        # Define decision variables
        x1 = LpVariable(name="x1", lowBound=0) # Quantity of product A
        x2 = LpVariable(name="x2", lowBound=0) # Quantity of product B

        # Define the objective function
        model += 3 * x1 + 2 * x2, "Objective"

        # Define constraints
        model += 2 * x1 + x2 <= 1000, "Raw_Material_Constraint"
        model += 4 * x1 + 2 * x2 <= 2400, "Labor_Constraint"
        model += 3 * x1 + 2 * x2 <= 1500, "Machine_Constraint"

        # Solve the linear programming problem
        model.solve()

        # Display the results
        print("Optimal Solution:")
        print(f"Quantity of Product A (x1): {x1.varValue}")
        print(f"Quantity of Product B (x2): {x2.varValue}")
        print(f"Maximum Profit (Z): {model.objective.value()}")
```

```
Optimal Solution:
Quantity of Product A (x1): 0.0
Quantity of Product B (x2): 750.0
Maximum Profit (Z): 1500.0
```

Warehouse Optimization Problem

A warehouse has three types of products, X, Y, and Z. The goal is to maximize the total revenue while considering storage space and labor constraints.

Decision Variables:

Let

- x be the quantity of product X to stock.
- y be the quantity of product Y to stock.
- z be the quantity of product Z to stock.

Objective Function:

Maximize the total revenue function:

$$Z = 10x + 15y + 8z$$

Constraints:

1. Storage Space Constraint:

The total storage space available is limited to 200 units, expressed as:

$$2x + 3y + 1z \leq 200$$

2. Labor Constraint:

Labor hours are limited to 150, expressed as:

$$4x + 2y + 3z \leq 150$$

3. Product X Availability:

The quantity of product X should not exceed 50 units, expressed as:

$$x \leq 50$$

4. Non-negativity Constraint:

The stock quantities cannot be negative:

$$x, y, z \geq 0$$

Python Code

```
In [2]: from pulp import LpProblem, LpMaximize, LpVariable

# Create a linear programming problem
model = LpProblem(name="Warehouse_Optimization", sense=LpMaximize)

# Define decision variables
x = LpVariable(name="x", lowBound=0) # Quantity of product X
y = LpVariable(name="y", lowBound=0) # Quantity of product Y
z = LpVariable(name="z", lowBound=0) # Quantity of product Z

# Define the objective function
model += 10 * x + 15 * y + 8 * z, "Objective"

# Define constraints
model += 2 * x + 3 * y + 1 * z <= 200, "Storage_Space_Constraint"
model += 4 * x + 2 * y + 3 * z <= 150, "Labor_Constraint"
model += x <= 50, "Product_X_Availability"

# Solve the linear programming problem
model.solve()

# Display the results
print("Optimal Solution:")
print(f"Quantity of Product X (x): {x.varValue}")
print(f"Quantity of Product Y (y): {y.varValue}")
print(f"Quantity of Product Z (z): {z.varValue}")
print(f"Maximum Total Revenue (Z): {model.objective.value()}")
```

```
Optimal Solution:
Quantity of Product X (x): 0.0
Quantity of Product Y (y): 64.285714
Quantity of Product Z (z): 7.1428571
Maximum Total Revenue (Z): 1021.4285668
```

Assignment.

A company produces two types of products, P1 and P2. The goal is to maximize profit while considering resource and demand constraints.

Decision Variables:

Let

- x_1 be the quantity of product P1 to produce.
- x_2 be the quantity of product P2 to produce.

Objective Function:

Maximize the profit function:

$$Z = 5x_1 + 3x_2$$

Constraints:

1. Raw Material Constraint:

The total raw material available is limited to 800 units, expressed as:

$$2x_1 + 4x_2 \leq 800$$

2. Labor Constraint:

Labor hours are limited to 200 hours, expressed as:

$$3x_1 + 2x_2 \leq 200$$

3. Demand Constraint:

The demand for product P1 is at least 100 units, expressed as:

$$x_1 \geq 100$$

4. Non-negativity Constraint:

The production quantities cannot be negative:

$$x_1, x_2 \geq 0$$